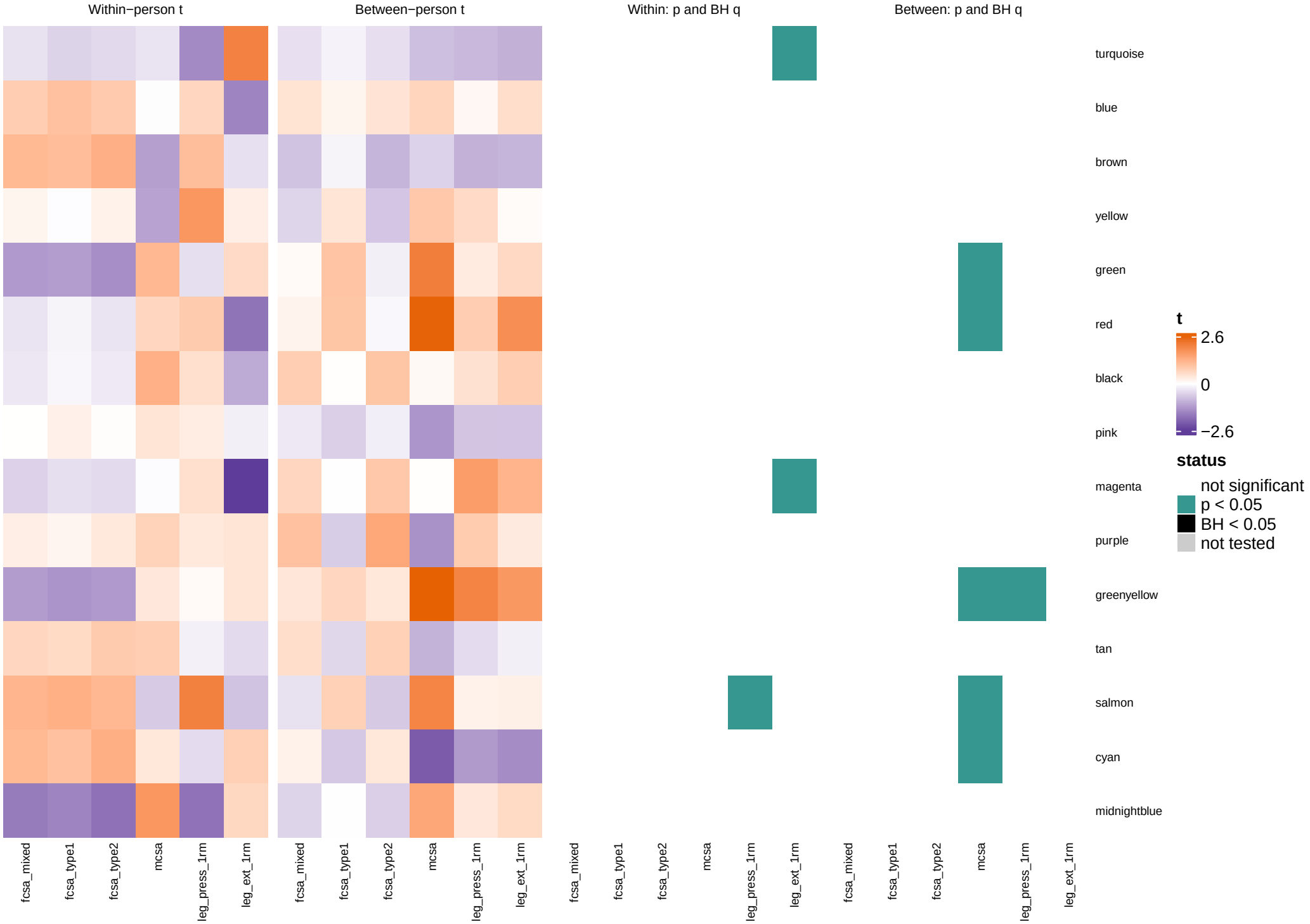
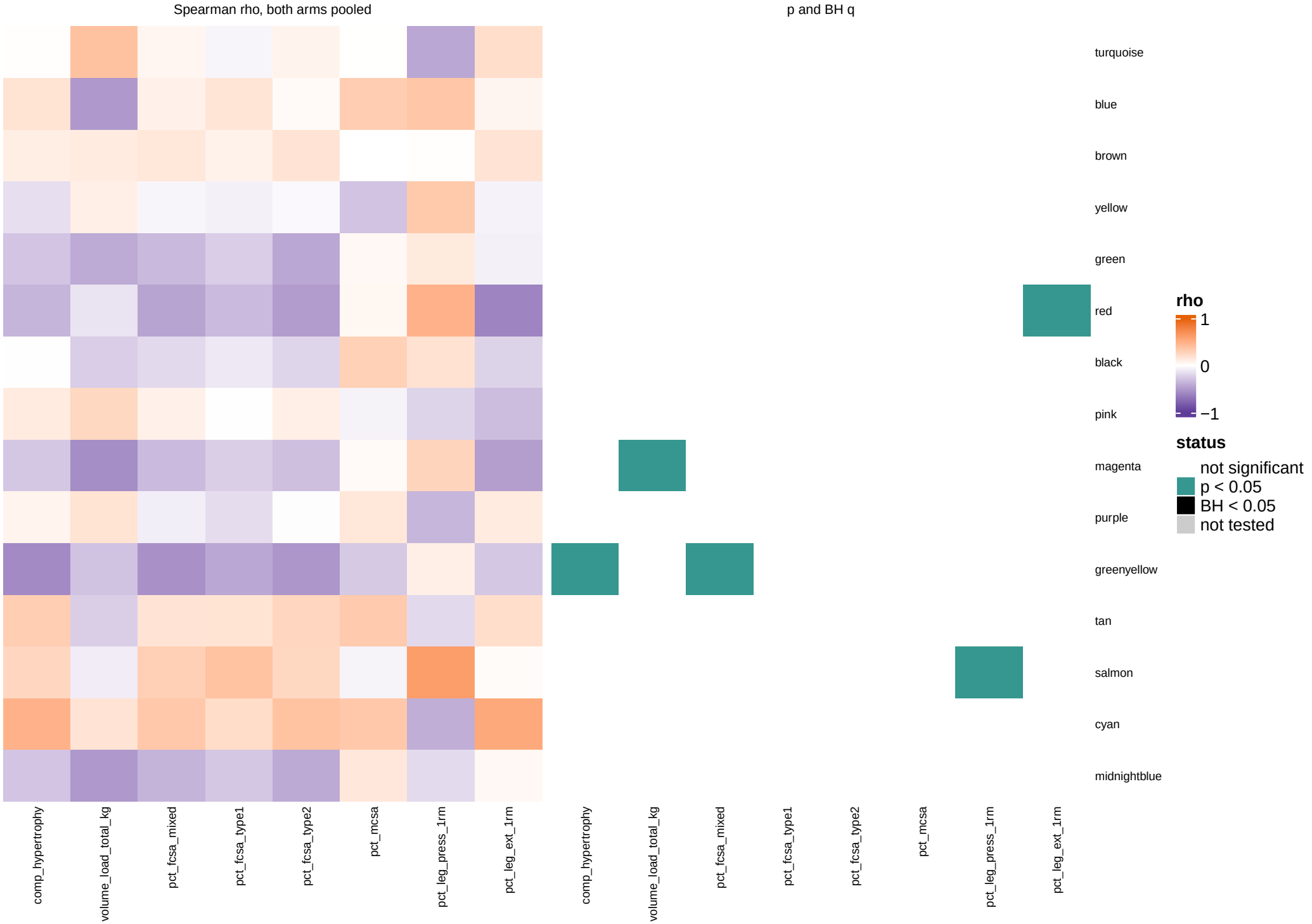
Sample model overview: modules



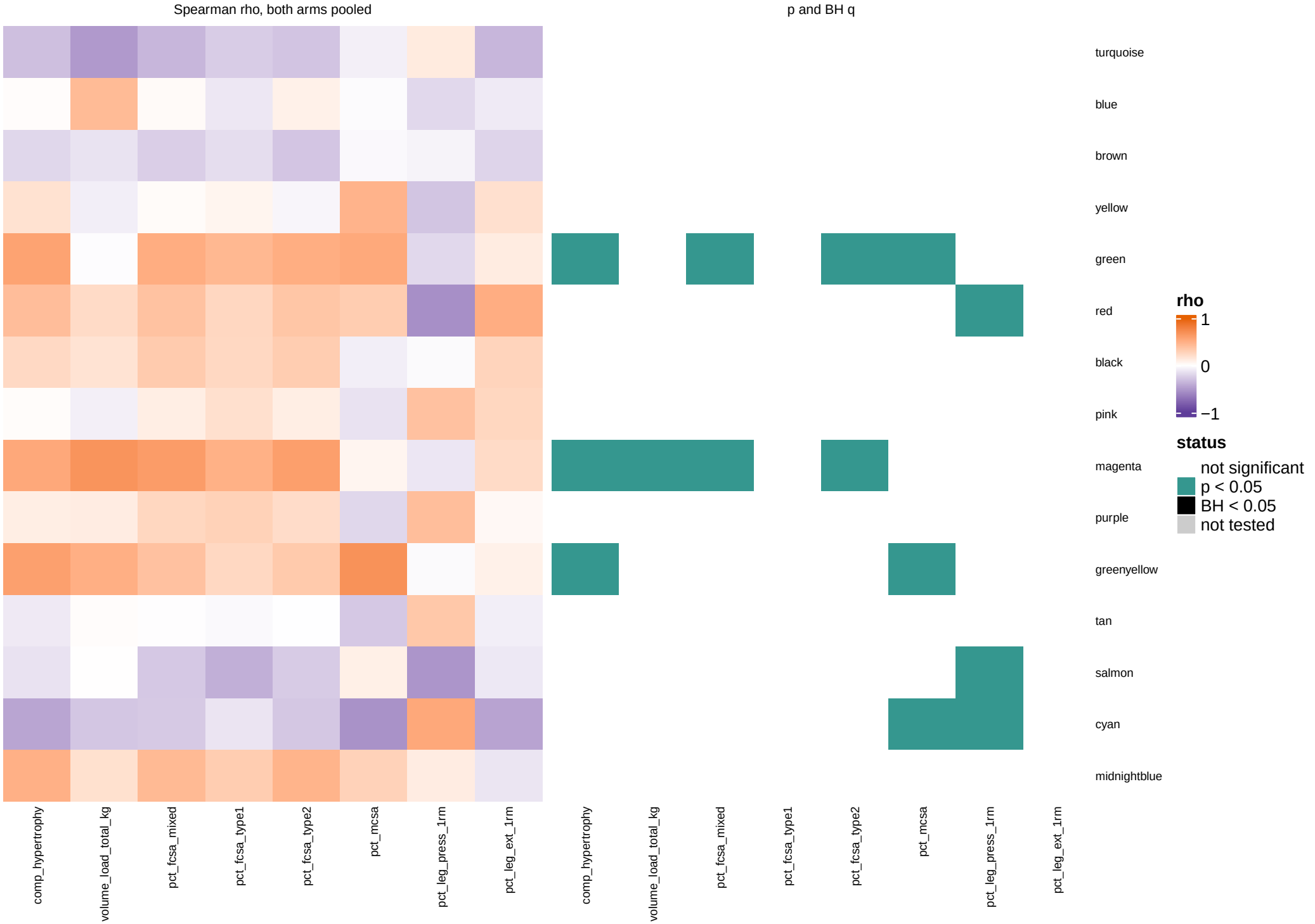
All 15 modules, one row each. Moderated t of the within and between terms, abundance ~ timepoint + between + within, subject blocked. Colours stop at the 99th percentile of |t|, 2.6. Rows follow module size. Grey: not tested. Table: c_data/04_associate.xlsx, model.

Change score overview, training window: modules



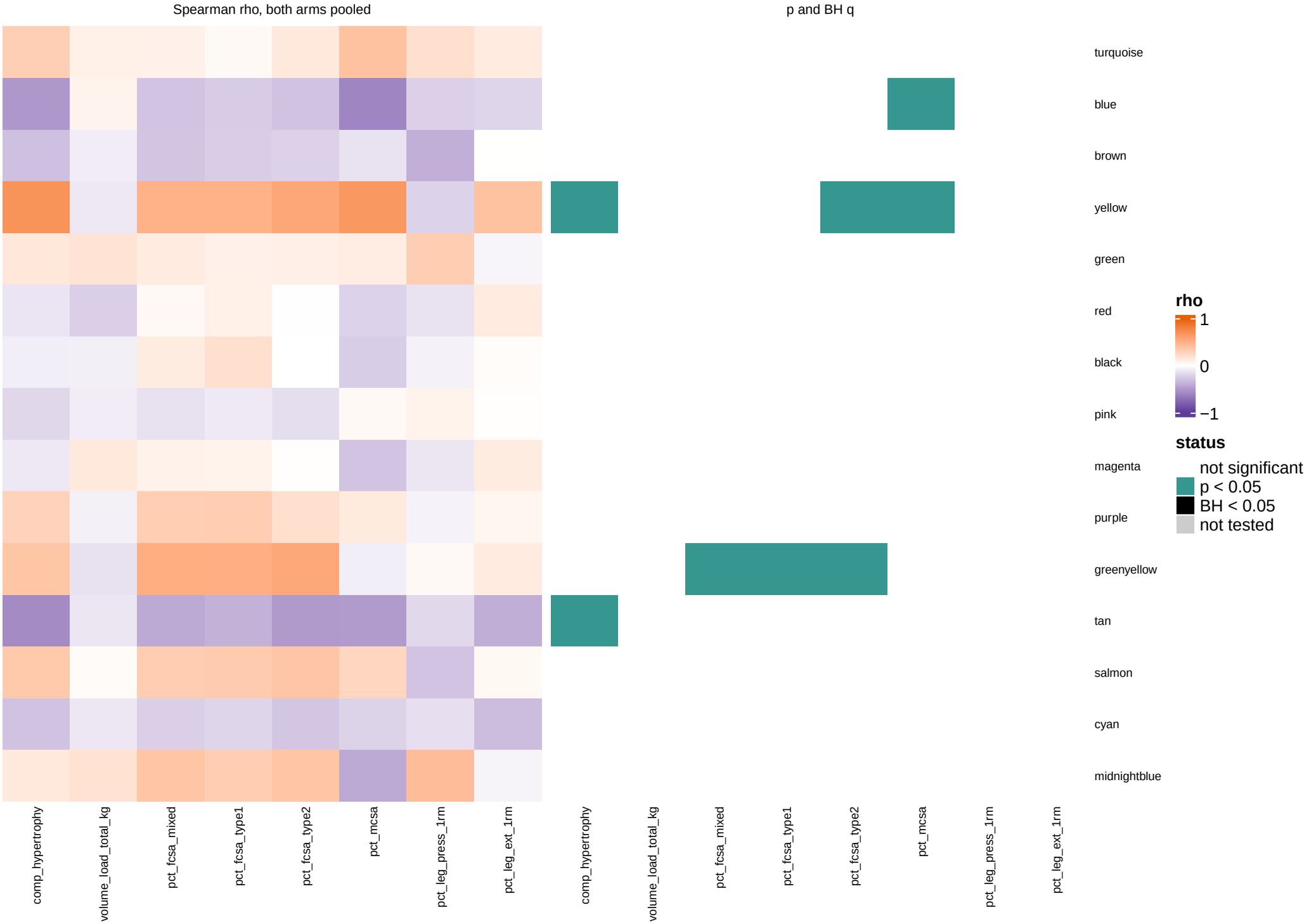
All 15 modules, one row each. Spearman rho between the eigengene (change, T1 to T2) and each outcome. Rows follow module size. Grey: not tested. Table: c_data/04_associate.xlsx, change_score.

Change score overview, baseline window: modules



All 15 modules, one row each. Spearman rho between the eigengene (T1 level) and each outcome. Rows follow module size. Grey: not tested. Table: c_data/04_associate.xlsx, change_score.

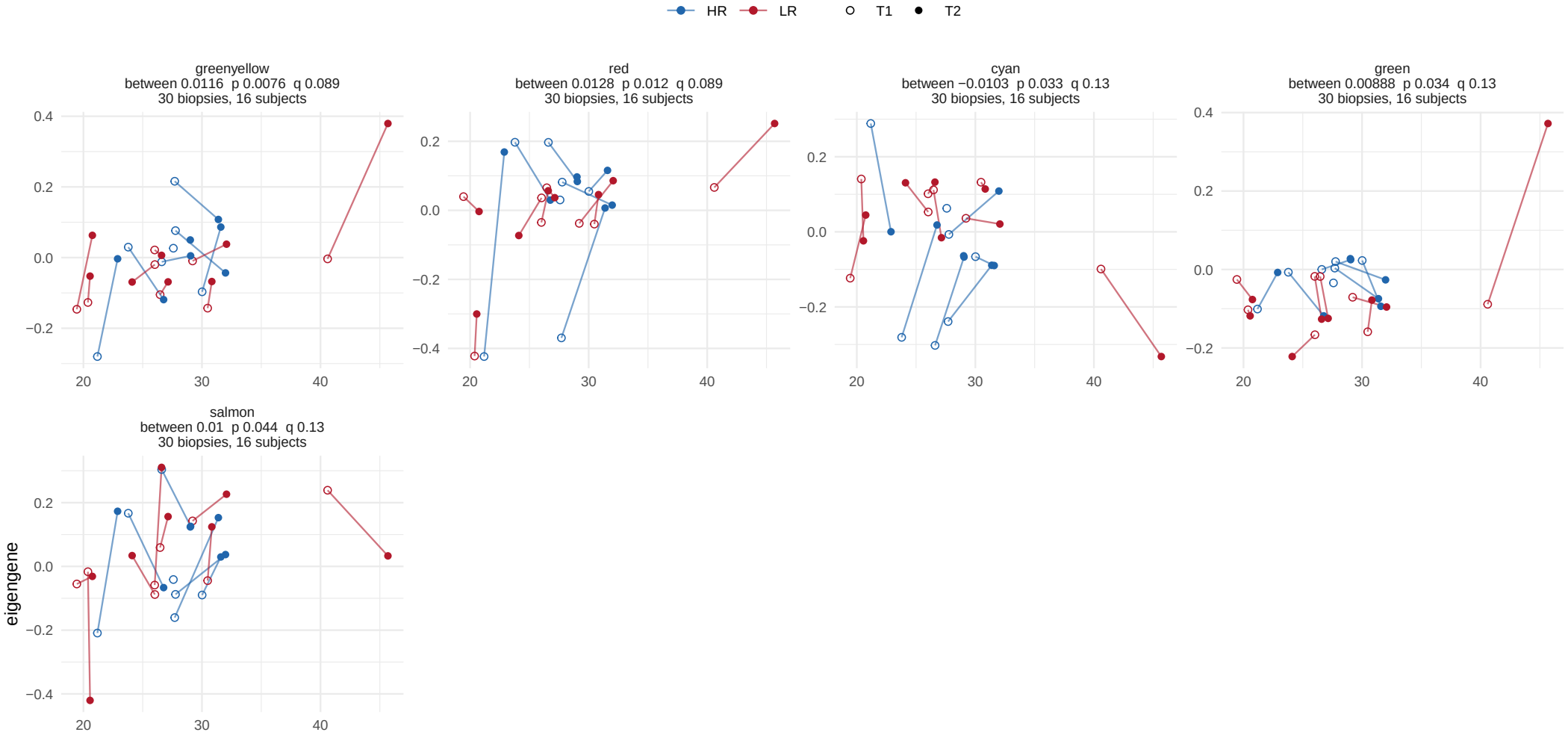
Change score overview, acute window: modules



All 15 modules, one row each. Spearman rho between the eigengene (change, T2 to T3) and each outcome. Rows follow module size. Grey: not tested. Table: c_data/04_associate.xlsx, change_score.

mcsa, between−person term

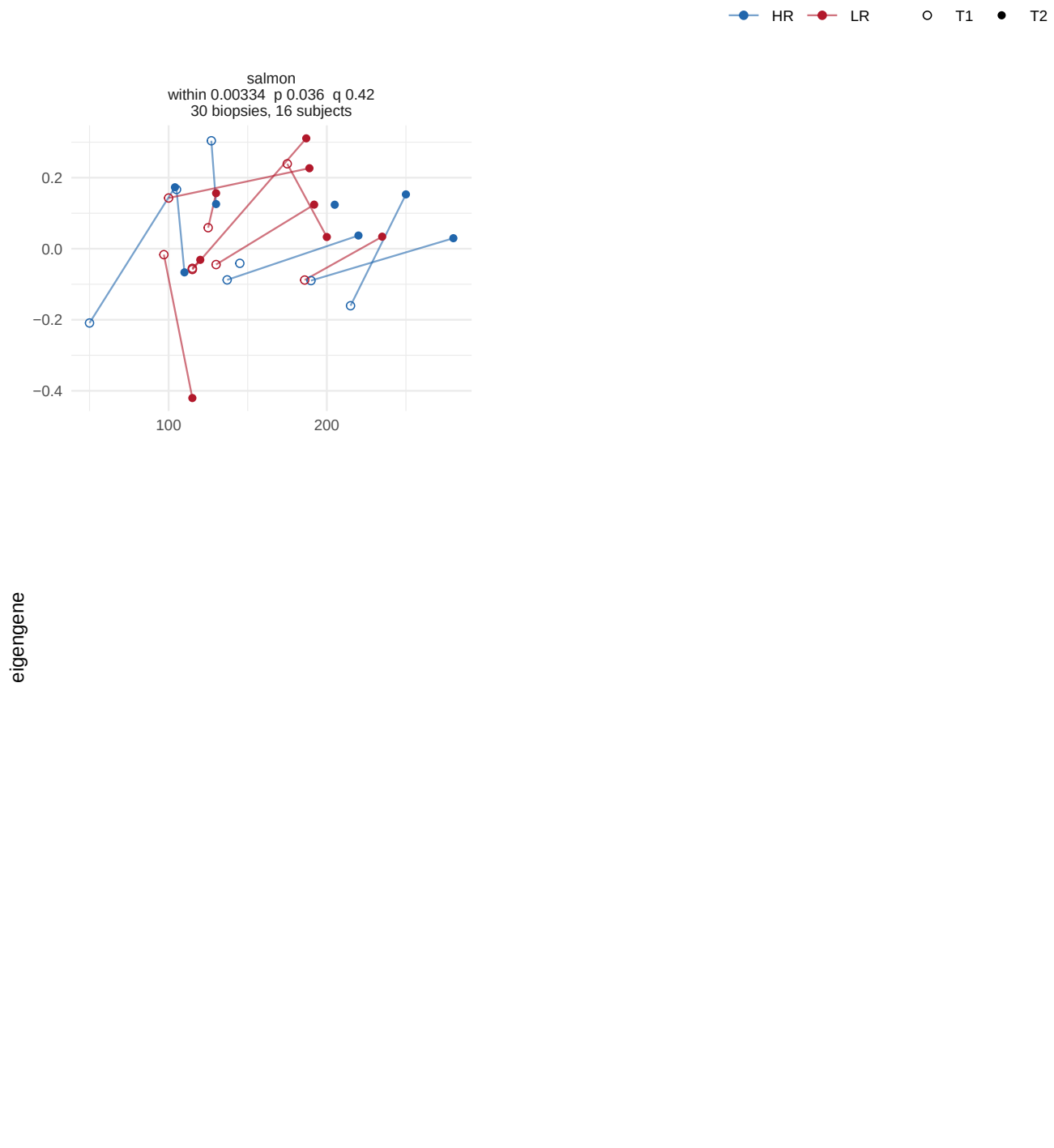
5 of 15 modules at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/04_associate.xlsx, model.

leg_press_1rm, within−person term

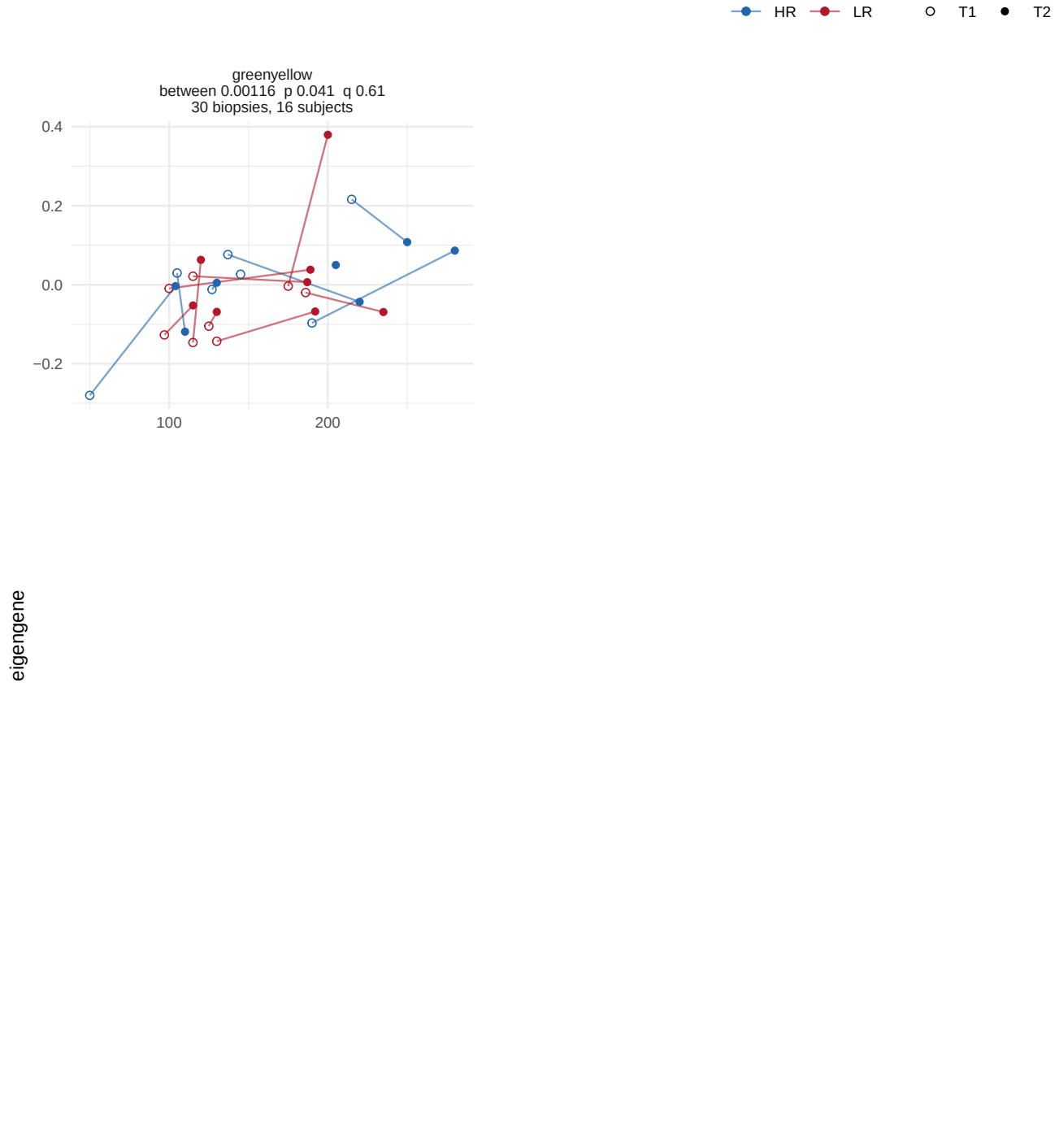
1 of 15 modules at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/04_associate.xlsx, model.

leg_press_1rm, between−person term

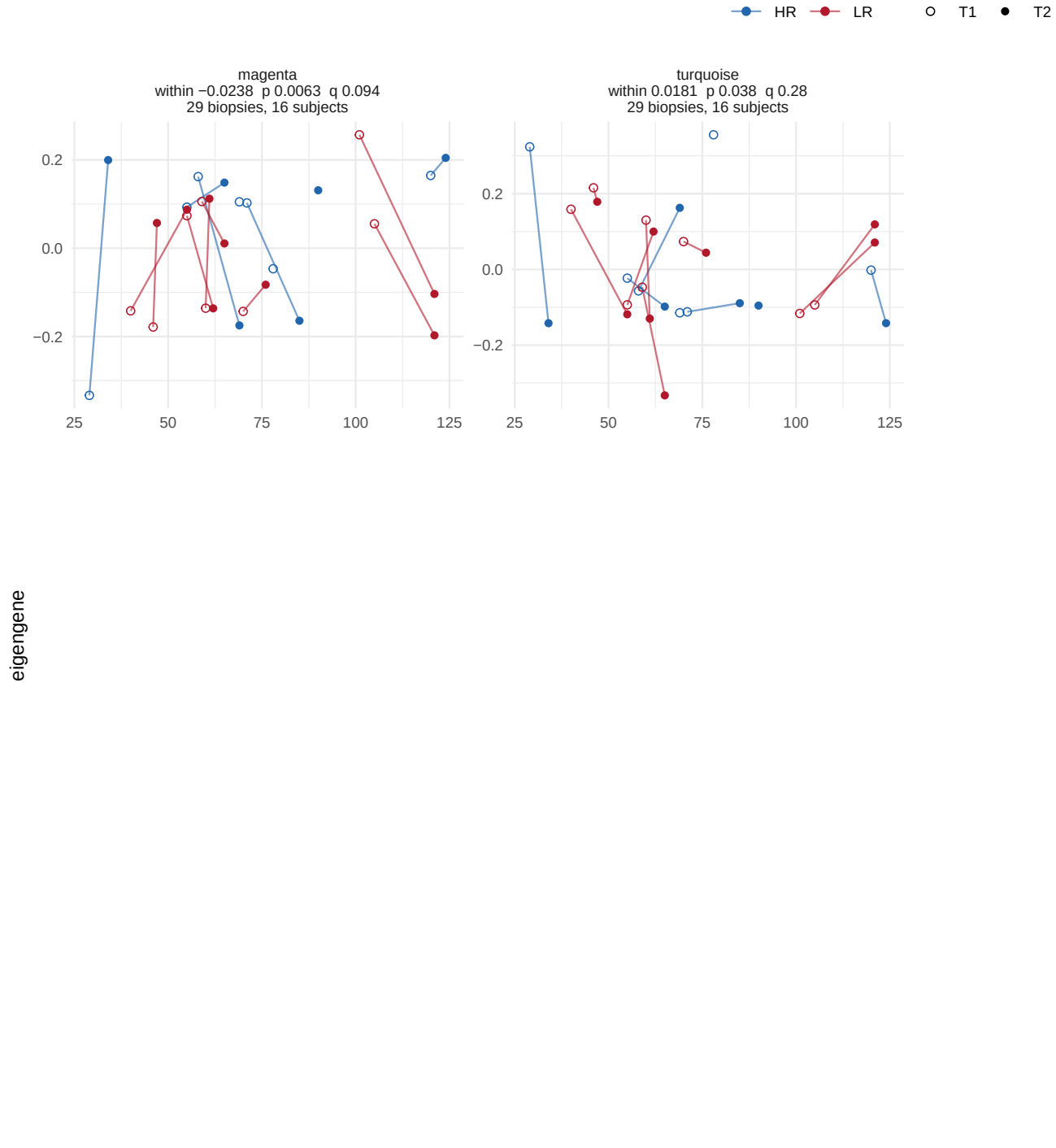
1 of 15 modules at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/04_associate.xlsx, model.

leg_ext_1rm, within−person term

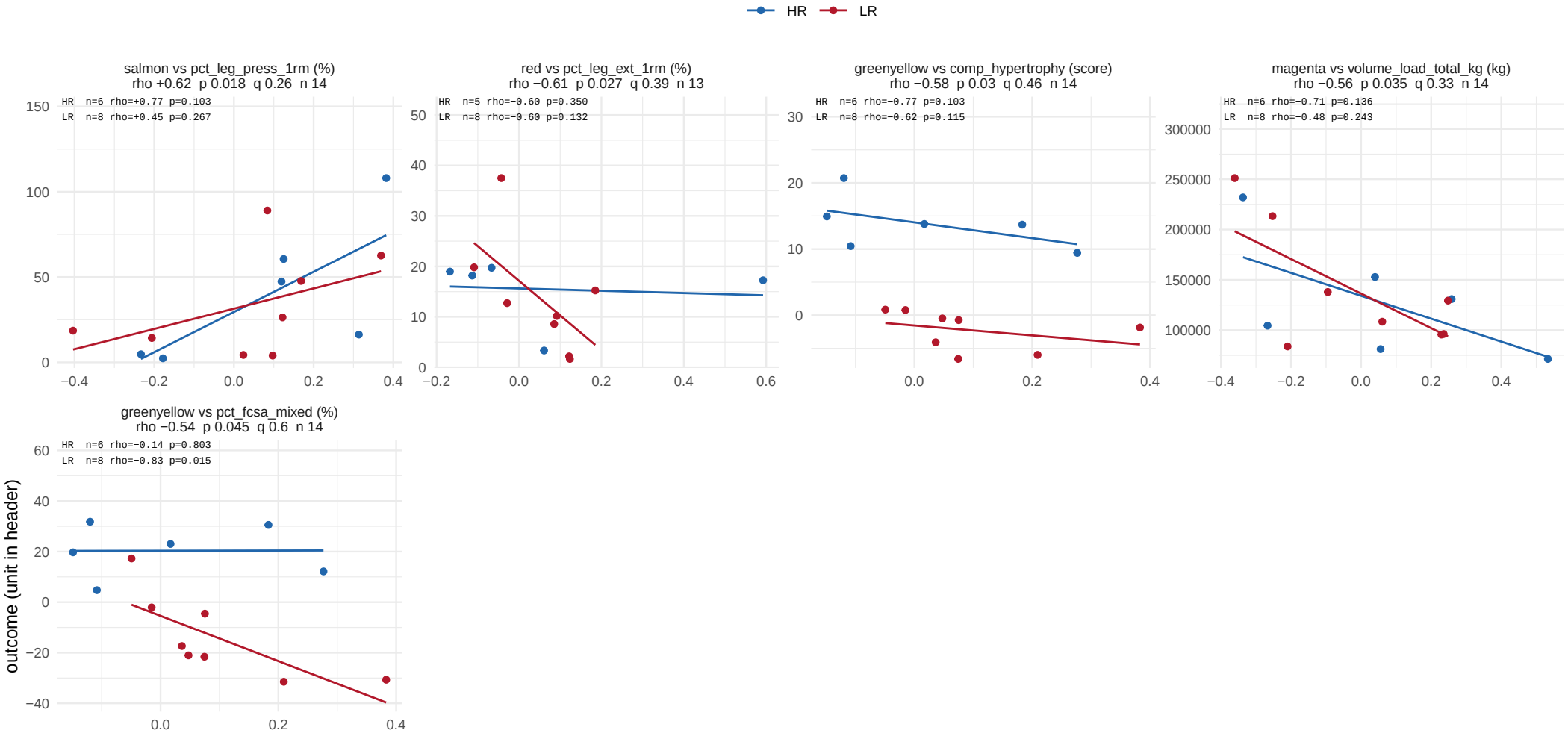
2 of 15 modules at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/04_associate.xlsx, model.

Change score, training window

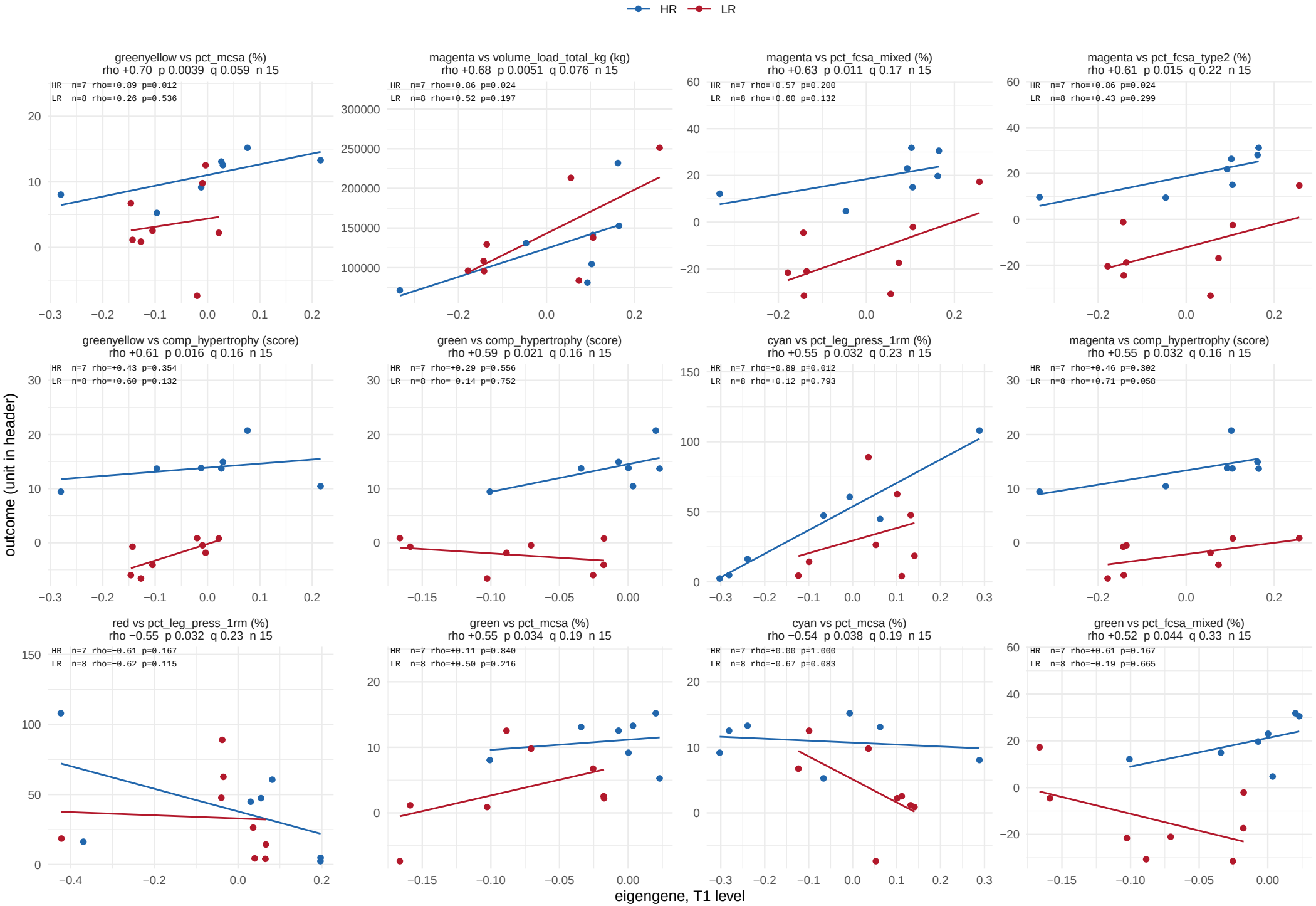
5 of 120 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



One point per subject. Lines are a least−squares fit per arm, drawn as a guide; the tests are Spearman. Header: rho and p pooled over both arms, BH q within window, outcome, and the subjects used. Corner: each arm's rho and exact p. Arm is the median split of comp_hypertrophy, so pooled correlations with it and with fibre area change partly carry the group split. Table: c_data/04_associate.xlsx, change_score and change_score_by_arm.

Change score, baseline window

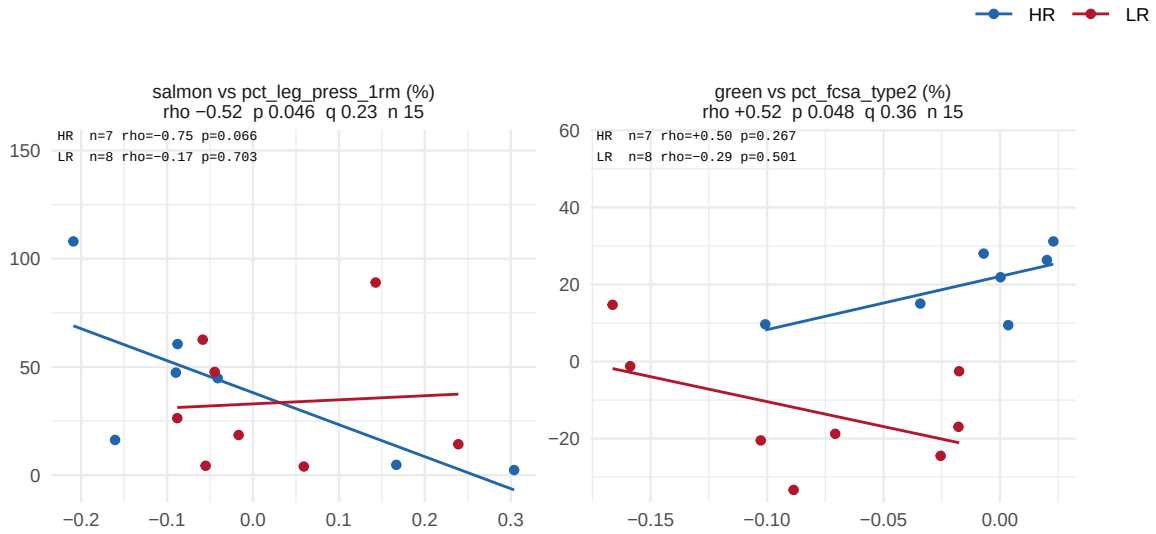
14 of 120 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



One point per subject. Lines are a least−squares fit per arm, drawn as a guide; the tests are Spearman. Header: rho and p pooled over both arms, BH q within window, outcome, and the subjects used. Corner: each arm's rho and exact p. Arm is the median split of comp_hypertrophy, so pooled correlations with it and with fibre area change partly carry the group split. Table: c_data/04_associate.xlsx, change_score and change_score_by_arm.

Change score, baseline window

14 of 120 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



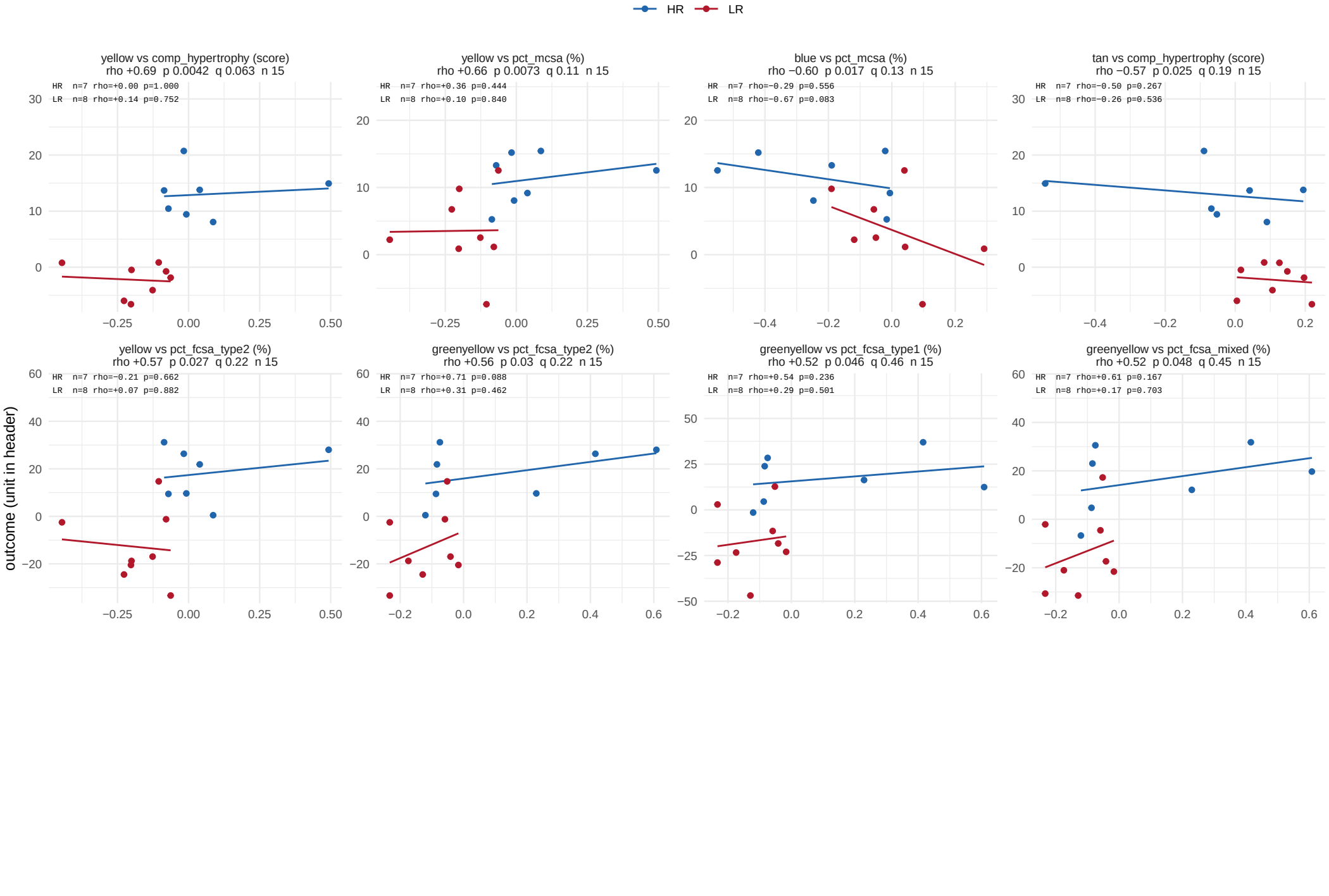
outcome (unit in header)

eigengene, T1 level

One point per subject. Lines are a least−squares fit per arm, drawn as a guide; the tests are Spearman. Header: rho and p pooled over both arms, BH q within window, outcome, and the subjects used. Corner: each arm's rho and exact p. Arm is the median split of comp_hypertrophy, so pooled correlations with it and with fibre area change partly carry the group split. Table: c_data/04_associate.xlsx, change_score and change_score_by_arm.

Change score, acute window

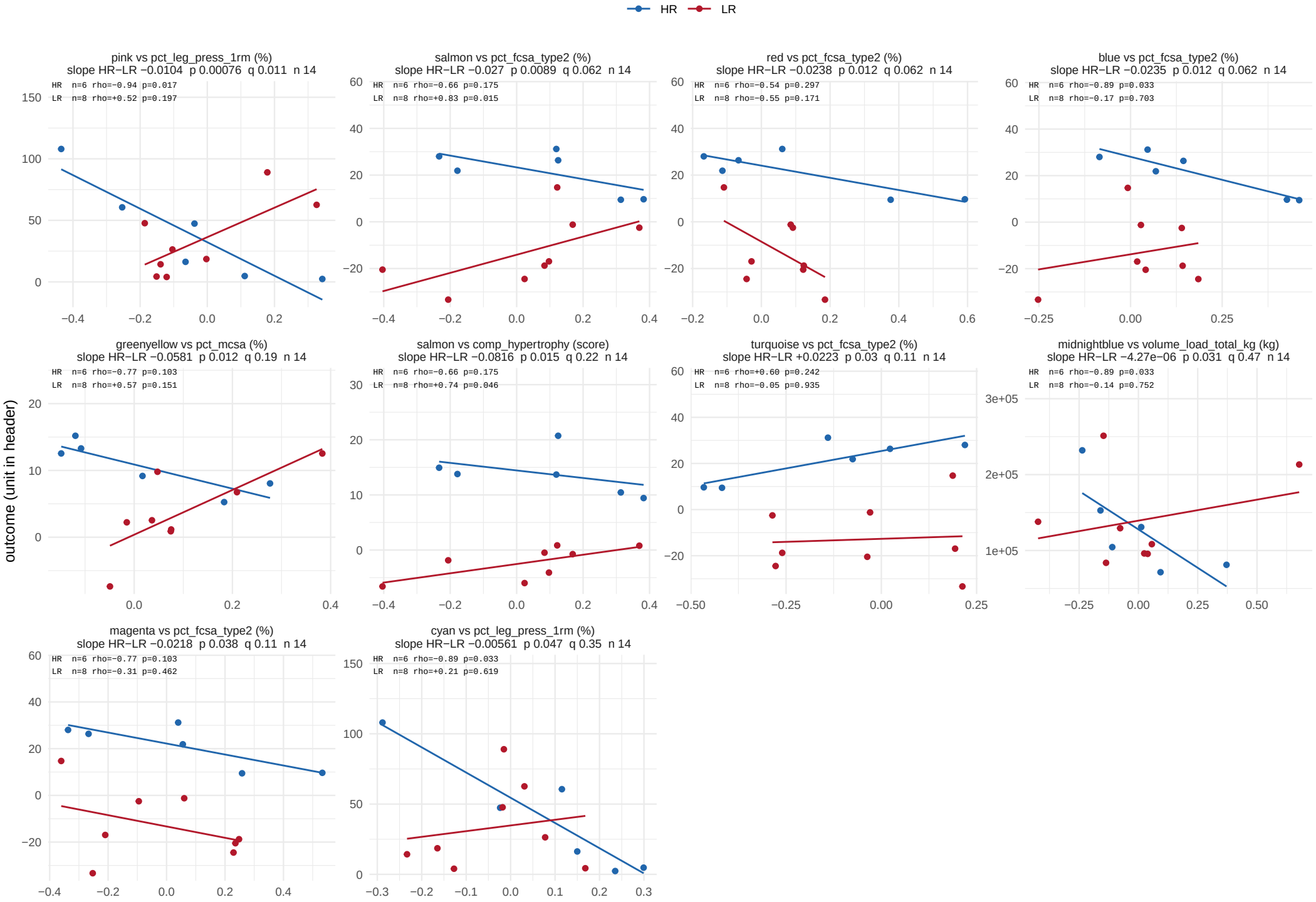
8 of 120 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



One point per subject. Lines are a least−squares fit per arm, drawn as a guide; the tests are Spearman. Header: rho and p pooled over both arms, BH q within window, outcome, and the subjects used. Corner: each arm's rho and exact p. Arm is the median split of comp_hypertrophy, so pooled correlations with it and with fibre area change partly carry the group split. Table: c_data/04_associate.xlsx, change_score and change_score_by_arm.

Arm difference, training window

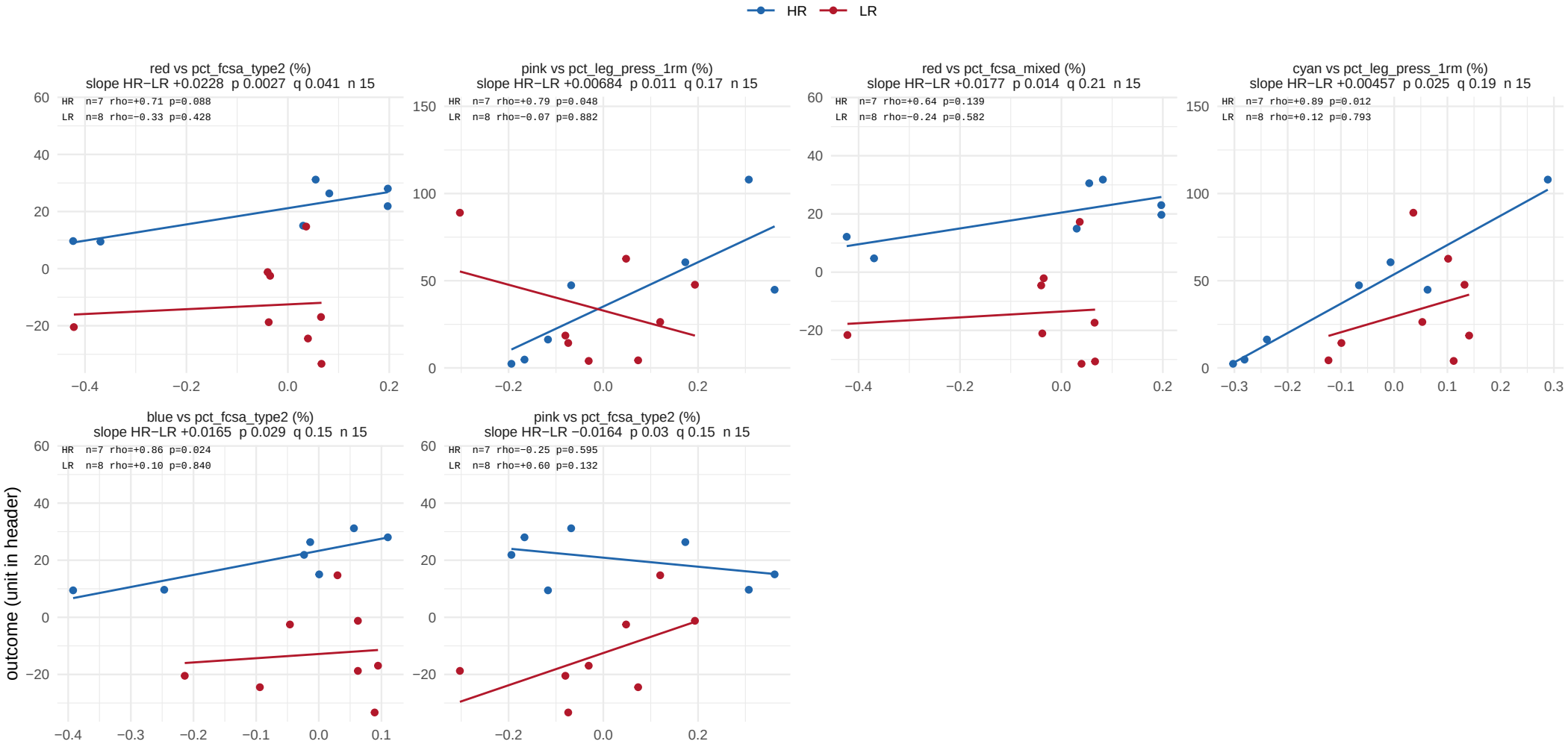
10 of 120 feature−outcome pairs at p < 0.05, 1 at BH < 0.05. Page 1 of 1.



One point per subject. Lines are a least−squares fit per arm. Header: HR's slope minus LR's per outcome unit from limma, feature ~ arm * outcome, with p, BH q within window, outcome, and the subjects used. Corner: each arm's Spearman rho and exact p. Table: c_data/04_associate.xlsx, arm_difference.

Arm difference, baseline window

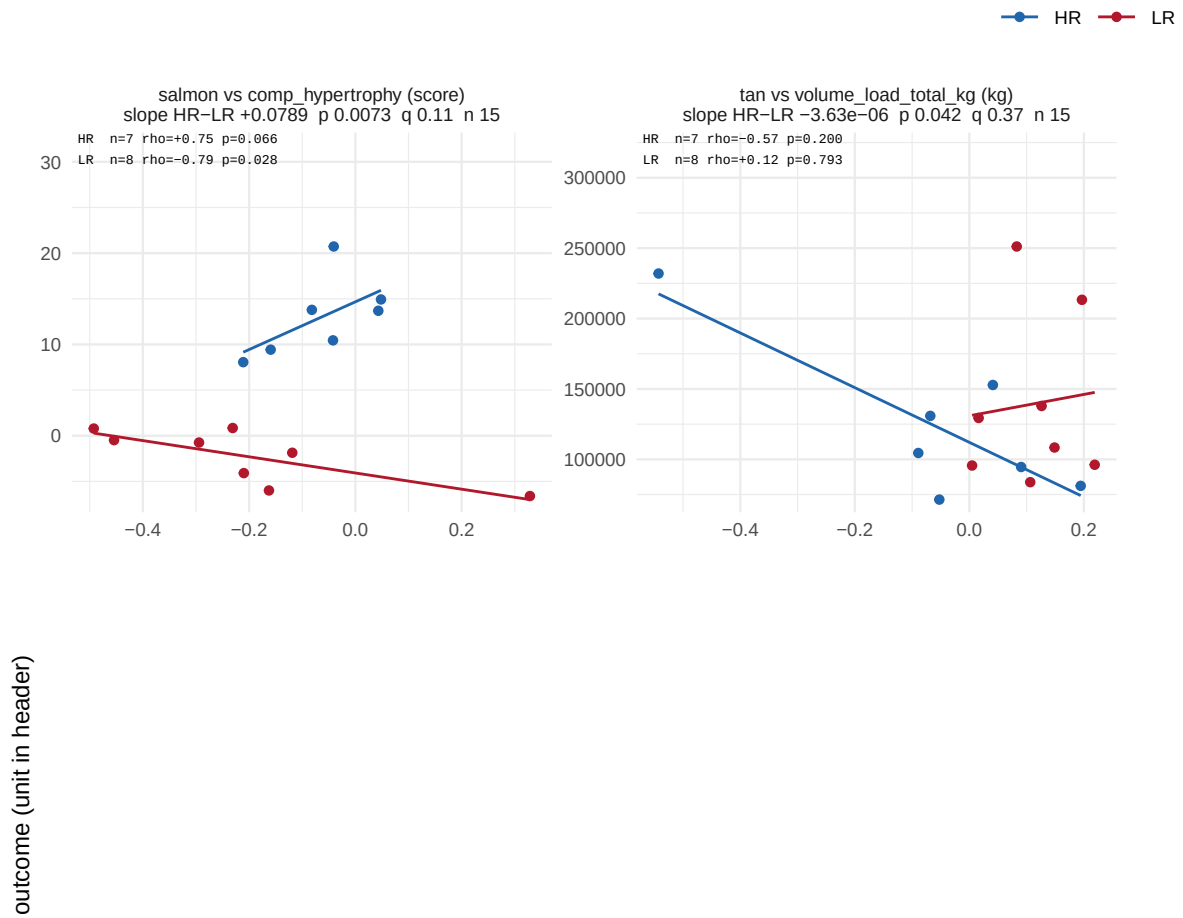
6 of 120 feature−outcome pairs at p < 0.05, 1 at BH < 0.05. Page 1 of 1.



One point per subject. Lines are a least−squares fit per arm. Header: HR's slope minus LR's per outcome unit from limma, feature ~ arm * outcome, with p, BH q within window, outcome, and the subjects used. Corner: each arm's Spearman rho and exact p. Table: c_data/04_associate.xlsx, arm_difference.

Arm difference, acute window

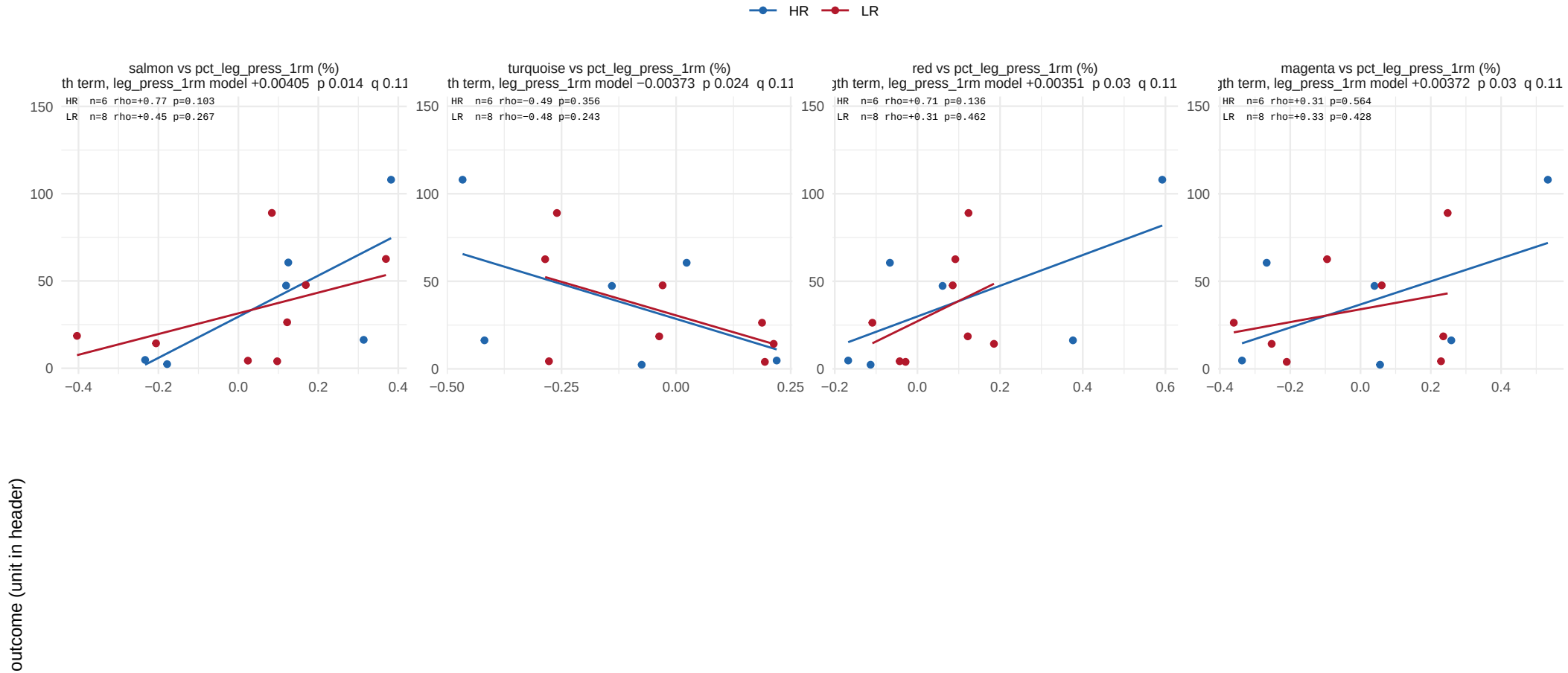
2 of 120 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



One point per subject. Lines are a least−squares fit per arm. Header: HR's slope minus LR's per outcome unit from limma, feature ~ arm * outcome, with p, BH q within window, outcome, and the subjects used. Corner: each arm's Spearman rho and exact p. Table: c_data/04_associate.xlsx, arm_difference.

Two predictor, training window

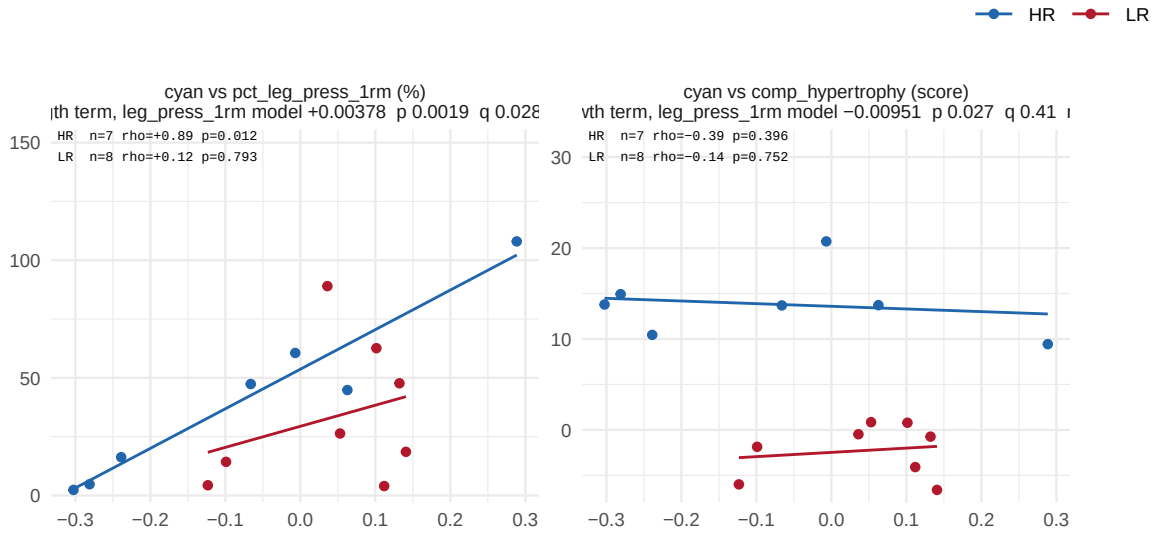
4 of 60 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Corner: each arm's Spearman rho and exact p. Table: c_data/04_associate.xlsx, two_predictor.

Two predictor, baseline window

2 of 60 feature−outcome pairs at p < 0.05, 1 at BH < 0.05. Page 1 of 1.



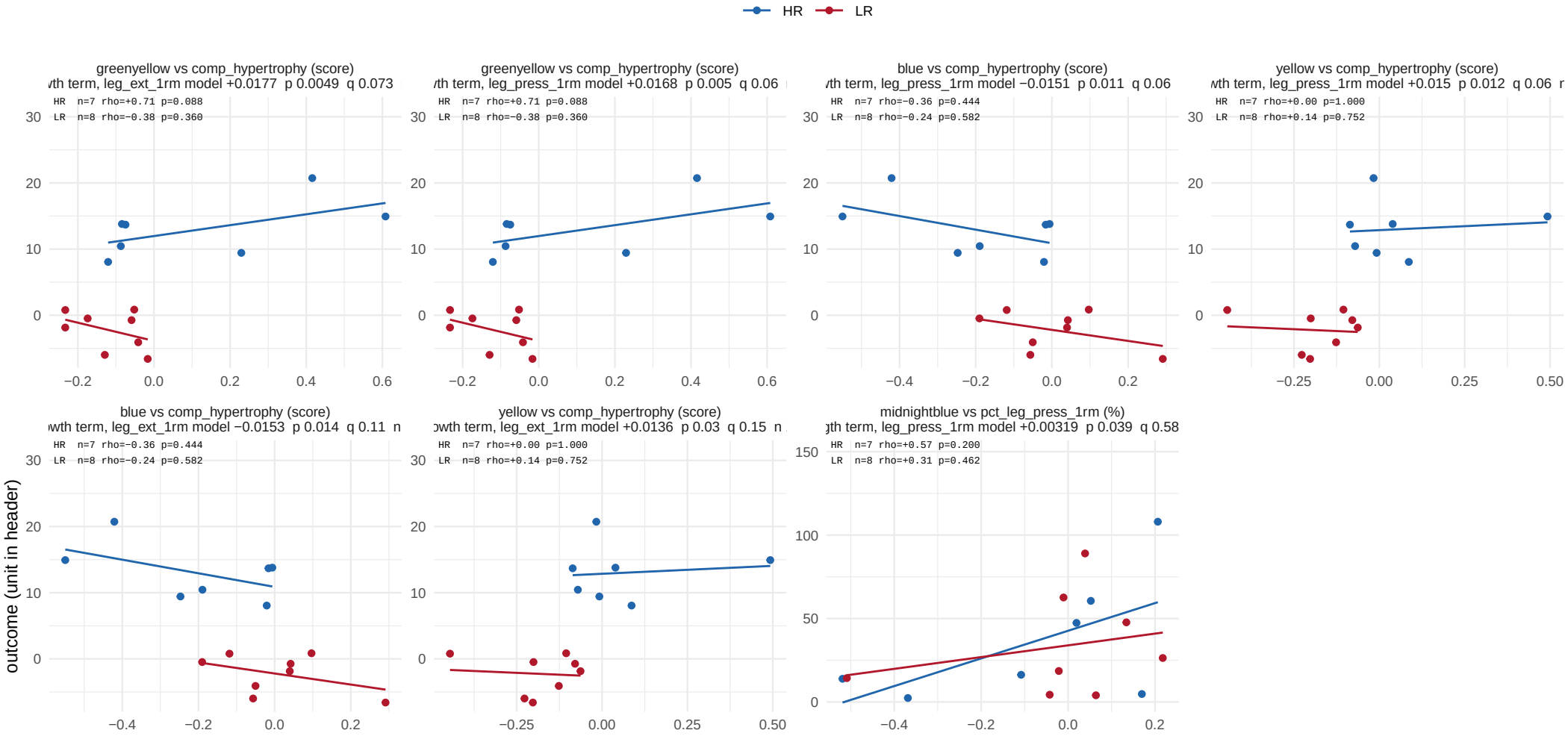
outcome (unit in header)

eigengene, T1 level

One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Corner: each arm's Spearman rho and exact p. Table: c_data/04_associate.xlsx, two_predictor.

Two predictor, acute window

7 of 60 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 1.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Corner: each arm's Spearman rho and exact p. Table: c_data/04_associate.xlsx, two_predictor.